**Cyber Security**
**Class No 09 (Lab)**
**Date: 05 Oct 2024**

**Topic: Latest Vulnerability Assessment and Information Gathering**

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**Steps:**

### 1. Understanding IP Records and Vulnerability Mapping

- **A Record:**
  - Maps a domain name to its corresponding IP address.
  - Example: 72.249.68.156 → juniv.edu.
- **Useful Tools for Domain and IP Analysis:**
  - **LeakIX.net:**
    - Monitor live vulnerabilities in systems.
    - View a graphical representation of active exploits.
  - **ThreatMap CheckPoint:**
    - Real-time attack protocol visualization.
- **Domain and IP Lookup:**
  - **Whois.com:** Identify the owner of a domain name and its registration details.
  - **Who.is:** Retrieve additional domain information.
  - **IPVoid:** Analyze details about the target IP address, including blacklists and geolocation.

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### 2. Vulnerability Scanning Tools

- **Nessus/Tenable Security Center:**
  - A leading vulnerability scanner.
  - Detects potential system weaknesses and matches them against policies.
  - **Installation:**
    - Install Nessus using a compatible hardware setup.
    - Activation Key Example: PN7W-Z9BP-KNM7-ZF5A-5NXP.
  - **Task:** Perform scans and generate reports on identified vulnerabilities.

- **Burp Suite Community Edition:**
  - Download from PortSwigger.
  - Used for web application security testing.
- **AbuseIPDB:**
  - Report malicious IP addresses and add them to a blacklist.
  - Visit AbuseIPDB to log suspicious activities.

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### 3. Advanced Information Gathering Techniques

- **Live Malware Analysis with Any.run:**
  - Platforms:
    - **Any.run:** Interactive threat analysis environment.
    - Malware reports: View Examples.
    - Submissions: Analyze Submissions.
    - Task-specific analysis: Task Report.
- **Service-Level Agreement (SLA):**
  - Defines service expectations between providers and users.

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### 4. Practical Demonstrations

1. **Checking Domain Vulnerabilities:**
   - Use tools like **LeakIX** and **ThreatMap CheckPoint** to identify and visualize live attacks.
   - Example IP: 72.249.68.156.
   - Verify details using **Whois** and **IPVoid**.
2. **Performing a Nessus Scan:**
   - Install Nessus and set up a vulnerability scan.
   - Evaluate results against the organizational security policies.
3. **Blocking Malicious IPs:**
   - Submit the malicious IP to AbuseIPDB for blacklist consideration.
4. **Simulating Malware Analysis:**
   - Upload suspicious files to **Any.run**.

- Analyze behavior and generate comprehensive malware reports.

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**5. Key Takeaways**

- Vulnerability assessment requires understanding domain and IP configurations (e.g., A records).
- Tools like Nessus and Burp Suite play critical roles in identifying security flaws.
- Real-time monitoring via platforms such as ThreatMap and LeakIX enables proactive security measures.
- Information gathering is the foundation of both defensive and offensive security operations.

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**Assignment Suggestions for Students**

1. Research live vulnerabilities for a specific IP or domain using LeakIX.
2. Perform a Nessus vulnerability scan and report findings.
3. Use Any.run to analyze a suspicious file and provide a detailed report.
4. Submit a suspicious IP to AbuseIPDB and track its status.